5 INTEGRATION



Figure 5.1 Iceboating is a popular winter sport in parts of the northern United States and Europe. (credit: modification of work by Carter Brown, Flickr)

Chapter Outline

5.1 Approximating Areas
5.2 The Definite Integral
5.3 The Fundamental Theorem of Calculus
5.4 Integration Formulas and the Net Change Theorem
5.5 Substitution
5.6 Integrals Involving Exponential and Logarithmic Functions
5.7 Integrals Resulting in Inverse Trigonometric Functions

Introduction

Iceboats are a common sight on the lakes of Wisconsin and Minnesota on winter weekends. Iceboats are similar to sailboats, but they are fitted with runners, or “skates,” and are designed to run over the ice, rather than on water. Iceboats can move very quickly, and many ice boating enthusiasts are drawn to the sport because of the speed. Top iceboat racers can attain speeds up to five times the wind speed. If we know how fast an iceboat is moving, we can use integration to determine how far it travels. We revisit this question later in the chapter (see Example 5.27).

Determining distance from velocity is just one of many applications of integration. In fact, integrals are used in a wide variety of mechanical and physical applications. In this chapter, we first introduce the theory behind integration and use